

Hossein Shahabadi Farahani

ML / MLOps ENGINEER · AI SYSTEMS FROM FIRST PRINCIPLES TO PRODUCTION

h.sh.farahani1@gmail.com · farahani1.github.io · [LinkedIn](#) · [GitHub](#) · [Google Scholar](#)

SUMMARY

I build AI systems end to end: data, model, serving, and whatever device the thing has to run on. I came to machine learning through mechatronics in 2016, back when we still wrote backpropagation by hand, and I've spent the years since at MAPNA Group on industrial ML — anomaly detection for power plants, medical imaging, and a Persian voice assistant that runs on a 1 GB Raspberry Pi. I also look hard at the organization a system has to live in, because in my experience that is where most AI projects actually fail.

EXPERIENCE

Machine Learning / MLOps Engineer

MAPNA Digital (*MAPNA Group; formerly the Digitalization Center of MECO*)

2022 – Present

- **Persian voice assistant for smart homes (2024–2025)**. Designed and built the whole platform: wake-word and silence detection on a Raspberry Pi 3 (1 GB RAM), four containerized cloud services (speech-to-text, text classifier, model registry, and an orchestrator), and device control over D-Bus.
- Designed the per-household intent model: a frozen ParsBERT encoder with a single dense layer, trained at setup from one example per command that an LLM expands exactly once. The LLM never sits in the inference path. Each model is under 40 KB; the system reached 96% command recognition and a 2.85 s average end-to-end response.
- Served a Persian speech-to-text transformer through FastAPI/Uvicorn, containerized with Docker and deployed on a cloud server.
- Built and deployed a retrieval-augmented chatbot: classifier training, model storage, retrieval, and the database, all running as services on cloud servers.
- **Chest X-ray classification (2023)**. Built a deep-learning classifier with Grad-CAM heatmaps, so physicians could see which regions drove each prediction. Demoed as a proof of concept to hospitals, radiologists, and insurers.

AI Strategy, Pre-Sales & Organizational Change

MAPNA Digital (*MAPNA Group*)

2021 – Present

- **Collaboration 2.0 (2025–2026)**. Diagnosed why a ~100-person R&D division (IIoT, condition monitoring, platform software) kept producing work its adoption teams rejected. Designed an intervention built around the PRD-authoring process, with Confluence as the shared record, to make ownership and contribution visible. Partly executed before the division's circumstances changed.
- Assessed organizational and technical readiness for AI in prospective contracts (wildlife monitoring, smart agriculture, smart city) and defined success metrics and validation strategies for them.
- Drafted roadmaps for the company's AI services: AI-as-a-Service APIs, industrial condition monitoring, the smart-home assistant.
- Presented the group's digital products, in English, to prospective clients and international visitors, adapting each presentation to the client's actual problem and digital readiness.

Data Scientist, Condition Monitoring

MECO (*MAPNA Group*), with the APAC Research Group, KNTU

2018 – 2020

- Built a semi-supervised anomaly detection pipeline over hundreds of sensor streams in thermal power plants. It learns normal behavior and localizes deviations to specific sensors and subsystems, so operators know where to look.
- Built root cause analysis tools that were used to diagnose real failures in operating plants.
- Designed soft sensors for real-time, low-cost estimation of key process variables, and made them transferable between units with domain-adversarial networks (published in *Computers in Industry*).
- Contributed these services to *MapnaMind*, the group's IoT and analytics platform.

Thermal System Designer

MSA (Machine Sazi Arak), Boiler Department

2015 – 2016

- Simulated and designed a steam accumulation system for Takht-Jamshid Petrochemicals.

SKILLS

- **Machine learning:** TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, LLM APIs; anomaly detection, root cause analysis, transfer learning and domain adaptation, explainability (Grad-CAM), speech and text classification, probabilistic models
- **Serving & backend:** Python, FastAPI, Uvicorn, Docker, HTTP APIs, D-Bus (Linux IPC), Raspberry Pi / edge Linux
- **Engineering practice:** Git, pytest, test-driven development, OOP and design patterns, Poetry
- **Product & process:** PRDs, Confluence, Jira, UML; AI-readiness assessment, roadmap design
- **Currently learning:** Kubernetes, cloud infrastructure, observability (Prometheus, Grafana)
- **Languages:** Persian (native), English (professional working proficiency)

PUBLICATIONS

- “Domain Adversarial Neural Network Regression to Design Transferable Soft Sensor in a Power Plant.” **H. Shahabadi**, A. Fatehi, M. Aliyari, A. Nadali. *Computers in Industry* (Elsevier), 2021.
- “On the Application of Domain Adversarial Neural Network to Fault Detection and Isolation in Power Plants.” **H. Shahabadi**, A. Fatehi, M. Aliyari. *19th IEEE ICMLA*, Miami, 2020. [IEEE Xplore](#)
- “Transfer Learning in Condition Monitoring of Process Plants.” **H. Shahabadi**, A. Fatehi. *2nd BRIC Symposium*, McMaster University, 2022.
- “Between-Domain Instance Transition via Gibbs Sampling.” **H. Shahabadi**, A. Fatehi, M. Aliyari. *arXiv*, 2020.
- Peer reviewer for *Neurocomputing* (Elsevier).

TEACHING

- **Lab Instructor, Industrial Process Control Lab**, KNTU (2018). Undergraduates identified and designed controllers for several pilot plants.
- **Teaching Assistant**, KNTU (2018): *Fault Detection & Identification* (graduate) and *Industrial Control* (undergraduate), both with Prof. Mahdi Aliyari.

EDUCATION

MSc, Mechatronics Engineering

K. N. Toosi University of Technology (KNTU)

2016 – 2020

- Thesis: *Improving Gas-Turbine Health Monitoring Using Deep Domain-Adaptive Neural Networks*. Supervisors: Prof. Alireza Fatehi, Prof. Mahdi Aliyari.

BSc, Mechanical Engineering

University of Arak

2011 – 2015

- Thesis: sizing the pressure vessel and control valves of a steam accumulator from a numerical simulation of its dynamics.

References available on request.